

QUICK SUPERVISION NOTE

AJCR.review-plan and runs 121 / 122 / 123

Status review of the August 2026 AJCR-first rank-one recovery route

REVIEWED	SCOPE	MODE
7 Aug 2026	Roadmap + latest run reports + key commits	Supervision audit; no local Lean rerun

Executive verdict

Yes: the agents are following the intended mathematical route, and the work since Tuesday is mostly productive. The important correction is that the current Phase 3 / Phase 5 "blocked" state is not merely a scheduler label. Both runs 122 and 123 independently identify the same genuine structural gap: the missing arbitrary-affine, family-level native presentation and cartesian base-change package needed to turn pointwise rank-one data into the public `PicRankOneOpen / FibrePresented` interface. That shared producer should now become the single critical-path task.

Short reply you can send

Thanks - I had a look at the AJCR.review-plan, the current Phase 3/4/5 notes, the latest reports in runs 121/122/123, and the relevant commits. My reading is that the execution is following the plan well, and I would not change the mathematical route.

The main point I would clarify is the meaning of the present blockers. In Phase 3 and Phase 5 this is a real shared blocker, not just an item waiting in a queue: the agents have reached the same missing arbitrary-affine family-level presentation/base-change interface. Run 122 has the public rank-one locus side essentially kernel-clean; run 123 has completed a genuine separably-closed translated-cover feeder; but neither can honestly manufacture the missing `PicRankOneNativePresentation/FibrePresented` producer from fieldwise $h_0=1$ and $H^1=0$. I agree with their decision not to hide that gap behind a wrapper or switch to the high-degree/Galois fallback routes.

So I would now concentrate one lane (probably the Phase-4/recovery lane, or a new lane with the right write scope) entirely on that family-level producer, and park 122/123 until it lands. Once that package is in place, the intended chain `PicRankOneOpen -> canonical inverse/rankOneAbelIso -> translated cover -> Pic^0` representability should start moving again. I would also encode this shared dependency explicitly in the roadmap, because Phase 3 and Phase 5 currently look independent in metadata even though their reports show they are blocked by the same theorem family.

Bottom line: keep the route; stop duplicating blocker discovery; give the shared family-level producer one clear owner and acceptance contract.

1. What I checked and what the current state says

The binding route is still the one you asked for: Div^g represented $\rightarrow \text{DivRankOneOpen} \rightarrow \text{rankOneAbelIso} \rightarrow \text{PicRankOneOpen} \rightarrow \text{translated cover} \rightarrow \text{pic0_sepClosed_representableBy} \rightarrow \text{finite-Galois descent} \rightarrow \text{pic0_representableBy} \rightarrow \text{JacobianData}$. The roadmap explicitly says the high-degree Abel quotient is fallback infrastructure only, and endpoint credit requires root reachability, an immediate consumer, a kernel build, an axiom audit, and deletion/specialization of obsolete paths. [R1]

The status picture

P	Roadmap item	State	Supervision note
0	Pinned baseline / critical root / route guard	DONE	Correctly closed. Baseline and stop rules are pinned.
1	Endgame contract / canonical curve spelling / anti-vacuity	DONE	Correctly closed. This is what prevents "fake completion" by wrapper lemmas.
2	Arbitrary-degree divisor spine	DONE	Correctly closed / capped.
3	One public rank-one Picard and divisor locus	BLOCKED	A real structural blocker remains: arbitrary-affine native presentation + base change. [R2]
4	Canonical family-level rank-one Abel isomorphism	ACTIVE	This is now the natural integration lane for the missing family-level producer.
5	Translated rank-one cover over a separably closed field	BLOCKED	The field/separable-closure feeder is real and landed, but endpoint membership still needs the Phase-3/4 producer. [R3,R6]
6	Pic^0 representability over the separable closure	PENDING	Correctly downstream; should not be forced before the rank-one family package.
7	Shared finite-Galois descent capstone	PENDING	Correctly downstream; no reason to activate the fallback/descent route early.
8	Construct JacobianData from pinned Pic^0 representation	PENDING	Correct endpoint.

The key interpretation of "blocked"

- Phase 3 is not blocked because the agent merely chose to work elsewhere. Its live roadmap contract asks for openness, arbitrary base change, and one public good-locus API; the current item is explicitly marked blocked. [R2]
- Run 122 says the gap is structural: the consumers already demand the exact arbitrary-affine native-presentation contract, but there is no lawful producer in the lane's writable boundary. [R5]
- Phase 5 says the same thing from the other side: the remaining blocker is a production arbitrary-affine presentation family with cartesian pushforward/base-change control, tied H_0/H_1 /rank data, and coherent native presentations on every affine pullback. [R3]
- Run 123 then stops for the right reason: producing this requires protected Phase 3/4 integration; a fieldwise substitute would violate the requested API contract. [R6]

Supervisory conclusion

The blocker is shared and substantive. The dashboard status is therefore informative: downstream feeder work has succeeded, but the proof architecture cannot cross from fibrewise/fieldwise rank-one witnesses to a family-level morphism of functors until the arbitrary-affine base-change package exists.

2. Run-by-run review: 121, 122, 123

Run 121 - rank-one recovery / integration lane

- Latest report: Phase 0-2 and the interface phase are pinned; Phase 4 is active; there is no fallback trigger. [R4]
- It identifies the exact next consumer as the canonical arbitrary-test divisor construction, while again naming the missing honest arbitrary-affine native presentation/base-change producer. [R4]
- Commit 7decac6 roots the conditional divisor-open representer and Pic^0 translated-layer feeder in the critical path, and explicitly records that the arbitrary-affine producer, family divisor/Abel inverse, and translated-cover endpoint remain gaps. Its reported verification is a full $\text{Pic}^0\text{CriticalPath}$ build with only the standard audited axioms. [C3]

Assessment of run 121

This is the best candidate to become the integration owner. It should now spend almost all effort on one theorem family that closes an immediate consumer. Re-running broad audits has diminishing value unless the API contract changes.

Run 122 - rank-one openness lane

- The public PicRankOneOpen / DivRankOneOpen side, arbitrary base-change interfaces, and inverse-facing consumers have been audited as kernel-clean in the run reports/screenshots.
- Commit 5ef9430 adds a native module bridge for glued datum; this is real infrastructure, not bookkeeping. [C1]
- The latest report refuses to introduce a vacuous wrapper when no lawful arbitrary-affine producer exists in its writable boundary. [R5]

Assessment of run 122

Correctly blocked. I would park this lane after a narrow handoff. Do not pay for repeated rediscovery of the same producer gap. Reactivate it as soon as the family-level producer lands, so it can close openness/base-change acceptance and the canonical inverse consumer.

Run 123 - separably-closed translated-cover lane

- This lane has produced genuine mathematical content. Commit 1759ffa adds the Pic^0 specialization of the translated-cover construction (162 lines) and keeps translator degree, finite support, residue degree one, and the descended-class/baseSubtraction compatibility tied to the same result. [C2]
- Commit 42b53cf registers five corresponding kernel-checked declarations as `lean_ok` graph nodes. [C4]
- The latest report says the Phase-5 feeder is complete, narrow builds passed for the relevant modules, and the audit exposed only `propext`, `Classical.choice`, and `Quot.sound`. [R6]
- It also explicitly says this is not yet PicRankOneOpen membership and cannot honestly be upgraded from field-level $h_0=1/H_1=0$. [R6]

Assessment of run 123

Useful feeder work is complete. The agent stopped at exactly the right boundary. Further fieldwise translated-cover lemmas are unlikely to move the endpoint until the family-level producer is available; park the lane rather than letting it drift into an alternative proof route.

One subtle positive sign

The three lanes are independently converging on the same missing contract. That is usually evidence that the architecture is now exposing a genuine bottleneck rather than that the agents are confused. The mistake would be to leave the bottleneck implicit and let each lane spend cycles re-proving that it exists.

3. Why the blocker is mathematically the right one

At a mathematical level, the gap is exactly the passage from pointwise/fibrewise information to a family construction. **Knowing that a line bundle has $h_0 = 1$ and $H_1 = 0$ after passing to fields is not, by itself, enough to define the inverse Abel construction as a morphism on arbitrary test schemes.**

- To construct a relative effective divisor canonically from a family of line bundles, the pushforward must behave as a rank-one locally free object on the relevant locus, not merely have one-dimensional fibres.
- The construction must commute with pullback. In Lean terms, the cartesian pushforward/base-change morphisms needed by the public API have to be `IsIso` in the precise hypotheses used by the family.
- The H_0 rank-one statement and H_1 vanishing must be tied to the same native presentation and transported coherently under arbitrary affine base change.
- Only then does the canonical section/evaluation divisor vary functorially and supply the inverse to the Abel map on the rank-one open locus.
- Consequently, a fieldwise translated-cover witness is a valid feeder, but it cannot replace the family-level representability/open-locus theorem. This is precisely why `run 123` is correct to refuse that shortcut.

The missing package, stated as one critical-path contract

	Contract	Acceptance test
A	Arbitrary-affine producer	For every affine test object in the intended locus, produce <code>PicRankOneNativePresentation / FibrePresented</code> data from the actual family, not from an unrelated existential carrier.
B	Cartesian base change	Prove the pushforward comparison morphism is an <code>IsIso</code> for every cartesian pullback required by the public locus API.
C	Tied cohomology/rank certificate	Package $H_1 = 0$, H_0 finite-projective of rank one, and the native presentation so they refer to the same object and remain coherent after pullback.
D	Canonical divisor consumer	Use the presentation/evaluation section to construct the arbitrary-test divisor and prove its base-change compatibility.
E	Close the public interface	Make <code>PicRankOneOpen / DivRankOneOpen</code> and the canonical inverse compile from the producer, then close <code>rankOneAbelIso</code> .
F	Audit and root	Build the narrow critical path, run the axiom audit, root the declarations, and remove/specialize any obsolete parallel path before crediting the phase.

Stop rule for the next lane

Do not credit a theorem that merely restates the desired property as a hypothesis, chooses a fieldwise witness, or introduces an alternative carrier disconnected from the public `PicRankOneOpen` consumer. The next accepted commit should reduce the actual family-level gap and make an immediate downstream consumer compile.

4. Concrete coordination changes I would make now

A. Give the shared blocker one owner

1. Create a narrowly named task such as "arbitrary-affine rank-one presentation and cartesian base-change producer" with write access to the Phase 3/4 integration files.
2. Assign run 121 / Phase 4 to it, or start one new lane with that exact scope. The owner should be responsible for A-F on the previous page, not for the whole remaining Jacobian proof.
3. Park runs 122 and 123 once their handoffs are recorded. Reactivate them only when the producer lands or when they receive a genuinely independent, producer-facing sublemma.

B. Encode the dependency in Horizon metadata

This is the main coordination smell I would fix. The current Phase 3 and Phase 5 roadmap files both still show `depends_on: []`, even though their own summaries/reports identify the same missing producer. [R2,R3]

- Make the shared producer a first-class roadmap/task node.
- Add explicit dependency edges from Phase 3, Phase 4 where appropriate, and Phase 5 to that node.
- Use the comments/inbox for local failure memory, but let the graph encode the actual blocking relation so new sessions do not rediscover it from scratch.

C. Tighten the next checkpoint

- Checkpoint success should be binary: does an arbitrary-affine producer exist and is it consumed by the public rank-one locus/canonical inverse?
- Require a narrow build plus axiom audit at the consumer root, as the parent plan already demands. [R1]
- Do not use token count, number of commits, or new helper declarations as progress proxies. The current runs have already done enough exploratory/audit work to know the bottleneck.
- Keep high-degree Abel quotient and early Galois descent behind the fallback guard unless the rank-one route is formally shown impossible under the current library, not merely difficult.

D. What should happen immediately after the producer lands

- 1 Run 122 closes `PicRankOneOpen` / `DivRankOneOpen` openness and arbitrary base-change acceptance.
- 2 Run 121 uses the same producer to finish the canonical arbitrary-test divisor and family-level `rankOneAbelIso`.
- 3 Run 123 upgrades its already-landed translated-cover feeder through the actual public rank-one interface, instead of a fieldwise substitute.
- 4 Only then activate Phase 6 (Pic^0 representability over the separable closure), followed by the shared finite-Galois descent capstone and `JacobianData` assembly.

My recommendation in one sentence

No strategic reset is needed: turn the repeated Phase-3/5 blocker into one explicit Phase-4 integration task, stop the two downstream lanes from spinning, and demand a consumer-closing arbitrary-affine base-change theorem family as the next milestone.

5. Dashboard snapshots supplied with the question

These screenshots are consistent with the repository state reviewed above. They are included only as visual context; the supervision conclusions rely primarily on the current roadmap/report/commit sources listed in Section 6.

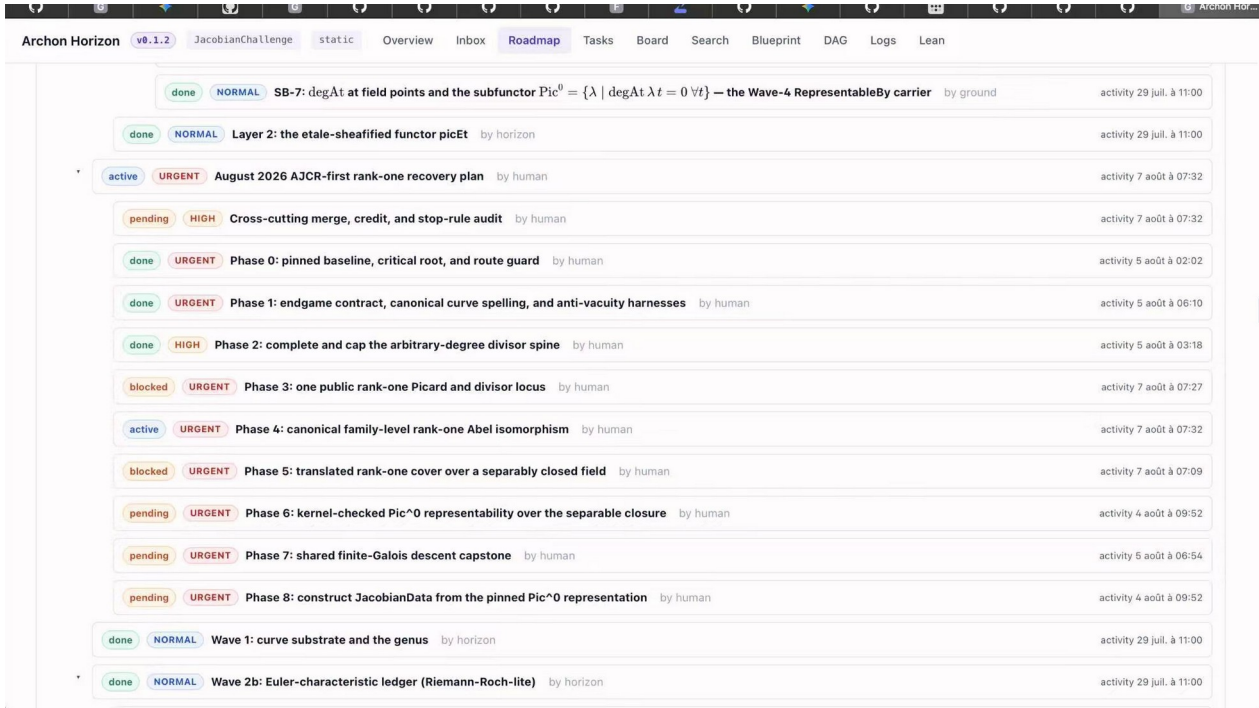


Figure 1. AJCR.review-plan dashboard: Phase 3 and Phase 5 blocked, Phase 4 active, Phases 6-8 pending.

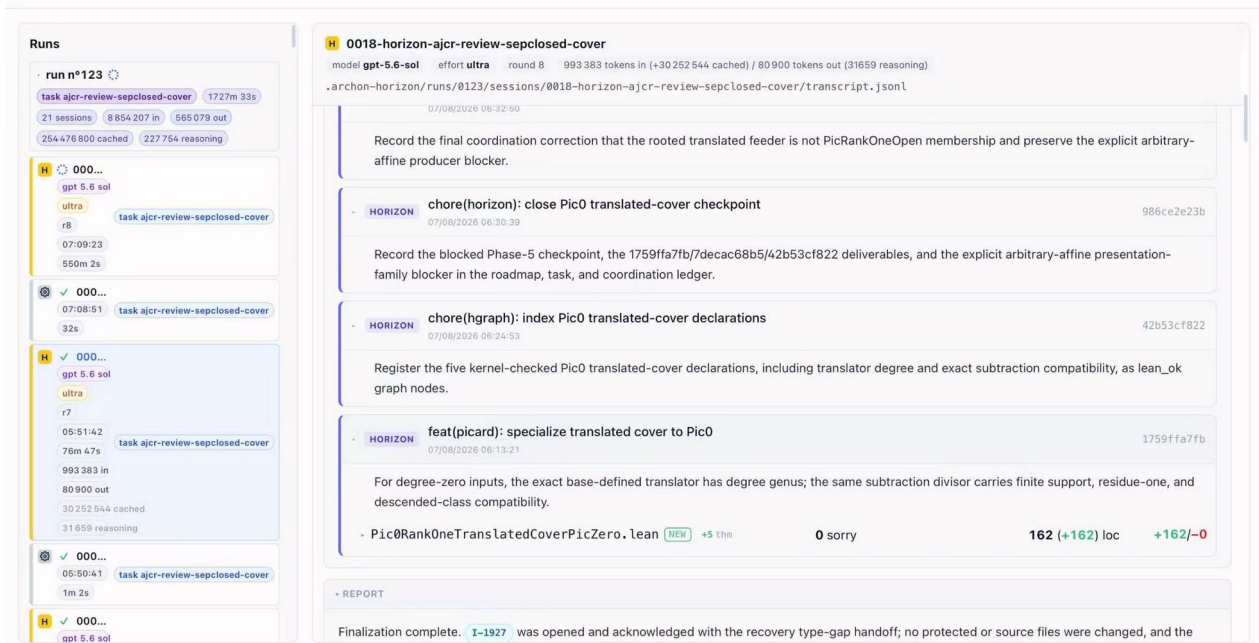


Figure 2. Run 123 commit/report view showing the Pic^0 translated-cover specialization and the explicitly preserved arbitrary-affine producer blocker.

6. Sources reviewed

Public repository/dashboard state was checked on 7 Aug 2026. Because the agents are still running, treat this memo as a point-in-time supervision audit. I did not independently execute Lean or lake build locally; build/axiom statements below are those recorded by the repository reports and commit summaries.

[R1] AJCR.review-plan - binding route and endpoint-credit rules -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/blob/main/.archon-horizon/roadmap/items/AJCR.review-plan.yaml> *Current plan file.*

[R2] Phase 3 - public rank-one Picard/divisor locus -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/blob/main/.archon-horizon/roadmap/items/AJCR.review-plan.p3-rank-one-loci.yaml> *Marked blocked; arbitrary base-change/public-locus acceptance.*

[R3] Phase 5 - separably-closed translated rank-one cover -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/blob/main/.archon-horizon/roadmap/items/AJCR.review-plan.p5-sepclosed-cover.yaml> *Feeder complete; production arbitrary-affine family remains blocker.*

[R4] Run 121 latest Horizon report (session 0028) -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/blob/main/.archon-horizon/runs/0121/sessions/0028-horizon-ajcr-review-rank-one-recovery/report.md> *Phase 4 active; same family-level producer identified.*

[R5] Run 122 latest Horizon report (session 0033) -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/blob/main/.archon-horizon/runs/0122/sessions/0033-horizon-ajcr-review-rank-one-openness/report.md> *Structural gap; refuses vacuous wrapper.*

[R6] Run 123 latest Horizon report (session 0026) -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/blob/main/.archon-horizon/runs/0123/sessions/0026-horizon-ajcr-review-sepclosed-cover/report.md> *Feeder completion, narrow builds/audit, explicit blocker and stop reason.*

[C1] Commit 5ef9430 - native module bridge for glued datum -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/commit/5ef9430cd9> *Run 122.*

[C2] Commit 1759ffa - specialize translated cover to Pic^0 -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/commit/1759ffa7fb> *Run 123; 162-line Pic^0 specialization.*

[C3] Commit 7decac6 - root rank-one openness and translated feeders -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/commit/7decac68b5> *Run 121; critical-path/root contract.*

[C4] Commit 42b53cf - index Pic^0 translated-cover declarations -

<https://github.com/AxelDlv00/LeanAlgebraicGeometry/commit/42b53cf822> *Run 123; five lean_ok graph nodes.*

[D1] Live Archon Horizon logs - <https://axeldlv00.github.io/LeanAlgebraicGeometry/#/logs> *Best live view while runs continue.*

[D2] Live Archon Horizon roadmap - <https://axeldlv00.github.io/LeanAlgebraicGeometry/#/roadmap> *Best live roadmap view while runs continue.*

[G] LeanAlgebraicGeometry repository README / methodology - <https://github.com/AxelDlv00/LeanAlgebraicGeometry> *Notes that roadmap/tasks/inbox record strategy and handoffs; graph frontier must be verified in tree.*

Final supervision call

Proceed. The route is intact and the agents are respecting the anti-shortcut constraints. The only intervention I would make now is organizational but mathematically decisive: concentrate ownership and compute on the shared arbitrary-affine presentation/base-change producer, then resume the already-prepared Phase 3 and Phase 5 consumers.